Akshay Borse

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PROFESSIONAL SUMMARY

Backend Engineer with 6+ years of experience building scalable, cloud-native systems. I've led high-impact projects—like an order optimization service that improved customer experience and cut operational costs by 25%. Skilled in microservices, infra-automation, and driving cross-team initiatives. Proactively led on-call reviews, design discussions, and knowledge-sharing sessions to improve system health and team alignment.

Beyond engineering, I'm driven by a curiosity for how cultures evolve, what helps societies thrive, and how we can use technology—especially in the age of AI—to reduce disparity and elevate human experience. I write about creativity, growth, and what it means to be human in a rapidly changing world.

EXPERIENCE

Senior Software Engineer | Microsoft | Redmond, WA | Sep-2025 to Present
(C#, .NET, Azure Functions, Cosmos DB, ARM, Bicep, Agentic AI, Agentic Risk, Microsoft Purview, Event Hub, Geneva, Kusto)

- Developing agentic risk detection capabilities in Microsoft Purview to identify and flag high-risk agents based on behavioral patterns and action history, enhancing data governance and insider threat detection.
- Implemented adaptive protection mechanisms in Microsoft Purview by enabling tenant-configurable risk models for digital agents, agentic users, and embodied agents—allowing granular classification and dynamic enforcement of security policies based on behavioral context.

Software Engineer 2 | Amazon | Seattle, WA | Aug 2019 – Aug 2025
(Java, Python, Rest, gRPC, Docker, ECS, Dynamo, S3, Lambda, Aurora, Step Function, App Mesh, Redis, Kibana, OpenSearch, Kinesis, CloudWatch, SageMaker, Bedrock, Anthropic's LLM, Prompt Engineering)

- Designed a microservice bootstrap library that standardized service creation across teams—cutting setup time from weeks to hours and saving 40–50 weeks. Leveraged custom DSL and enabled plug-and-play support for AWS infra using declarative config.
- Enhanced the bootstrap library for Multi-Service Deployment in a single pipeline, optimizing maintenance for a microservices architecture with 60+ services. Reduced the service count by 9, with plans in place to further streamline and drive operational excellence.
- Optimized Docker builds, cutting container image size by 83% and build time by over 50%, accelerating CI/CD and saving 60GB+ of storage.
- Established and led a logging/metrics optimization initiative and process during AWS migration—standardized logs, removed redundant code, cutting noise, improving debuggability, and reducing CloudWatch costs by $250K+/month.
- Built a Restock Alerting Service to notify customers when out-of-stock items were restocked increasing sales by 13% for cancelled ASIN's.
- Implemented the Just-In-Stock service on native AWS to track real-time inventory changes and auto-optimize fulfillment, increasing early delivery rates by 16% and improving supply chain responsiveness.
- Designed a self-service UI and backend to modify delivery dates for high-volume Amazon Business orders—eliminating 2–3 hours of manual ops per request and reducing dependency on external teams. Also mentored an intern during development.
- Engineered a Replay Service using a lightweight wrapper over AWS Step Functions to schedule event replays up to 1 year—eliminating cron-based scheduling and reducing operational overhead with built-in visibility and fault tolerance.
- Configured a dynamic log-level API for production, enabling real-time debugging without redeployments and cutting incident resolution time by over an hour—especially valuable in complex, distributed services.
- Led the design for a RAG-based service that leverages a model deployed on Amazon SageMaker to integrate real-time inventory signals and vector search, improving delivery predictions and driving a 7% increase in global sales.
- Redesigned 37 email templates by integrating an LLM via Amazon Bedrock, using a RAG-based approach with a curated corpus to constrain generation—ensuring compliance, improving personalization, and reinforcing brand consistency.
- Integrated a binary classification model (XGBoost) to predict potential order delays before SLA breach, enabling early delay alerts to customers and improving trust in delivery communication.
- Architected and deployed backend integration infrastructure for SageMaker hosted quantile regression model powering real-time delivery range predictions, driving statistically significant annualized ~28MM profit gain.
- Architected an event orchestration layer for AWS-based microservices, enabling real-time coordination across 100K+ TPS and 16 services—balancing REST and gRPC protocols to support 8 business-critical workflows.
- Coordinated end-to-end load testing for 60+ distributed services during high-traffic events (Prime Day, Black Friday), aligning with upstream and downstream service owners to proactively surface scaling issues and ensure fault tolerance under peak load.
- Developed a dry-run simulation tool for incoming orders, allowing non-engineering stakeholders to visualize system behavior—eliminating engineering dependencies, improving planning agility, and reducing deep-dive turnaround time from hours to minutes.
- Proactively led cultural and operational improvements by launching on-call reduction reviews, post-launch knowledge sharing, and design discussion sessions—reducing Sev2 alerts by 60%, improving team-wide system understanding, and strengthening long-term design thinking.
- Led TRA readiness for a Tier 1 service (CI/CD, multi-AZ, fault injection, canary), while building reusable tools to improve dev efficiency—automated UML sync to wiki, introduced reusable wiki headers, and reduced test setup complexity via Lugia integration.

Software Engineer | Liquiron | San Jose, CA | Dec-2018 to Jan-2019
(OAuth 2.0, Passport.js, Rest API, Vue.js, Google Firebase, Node.js, Mocha & Chai, Jest, Serverless Hosting, Stateless Backend)

- Streamlined backend authentication architecture using OAuth 2.0 and Passport.js, improving code modularity, security, and maintainability.
- Migrated 2 backend services to Google Firebase, enabling serverless deployment and eliminating local server dependencies.
- Designed 4 Rest APIs to support 10 front-end features in a SPA environment, optimizing request handling and reducing server load by 36%.

Software Engineer | Persistent Systems | Pune, India | Sept-2016 to Jul-2017
(CNN, AWS EC2, REST, NodeJS, JavaScript, Python, Axios, Quasar)

- Developed a real-time CNN classifier to classify videos and images by 7 sentiments, with 87% accuracy and efficient sentiment analysis.
- Streamlined website into a single-page application, boosting response time by 25% and simplifying testing and maintenance.
- Cached Fonts, JS, CSS, images using Google workbox thus improving the server response by 9% under unreliable networks.
- Improved a feedback system for reporting bugs & defects in new interface functionality, which was adopted by 6 teams across the organization.

EDUCATION

- Santa Clara University, Santa Clara, CA | Master's in Computer Science and Engineering. | Sep - 2017 to Jun-2019 | GPA - 3.77
- Pune Institute of Computer Technology (PICT), India | Bachelor's in Computer Science and Engineering. | Jun - 2012 to Jun-2016 | GPA - 3.34

PROJECTS - HTTPS://GITHUB.COM/AMSBORSE

- Hero Recommendation System for DotA 2 - Used 50,000 matches for training and 10,000 for testing. Used DotA-2 API to extract the data.
- Natural Language to SQL Query Generator - Independent of database schema. Accuracy 85% out of 10K test set.
- Face & Motion Recognition - PCA with L2-norm maximization, Face Accuracy – 99%. Edge detection, Keras CNN, Motion Accuracy - 77%.

ACHIEVEMENT

- Started my blog on improving creativity, growth and human experience – to help navigate the AI age with originality & depth.
- Received Just Do It Award for developing an Automated Service Creation tool that deploys micro services on AWS within an hour.
- Won a prize for the most innovative project at the 'Bronco Hackathon' at Santa Clara University.